DIGITAL FORENSICS LAB REPORT

Windows Registry Forensics Analysis

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Platform	INE - Incident Response & Digital Forensics
Lab	Windows Registry Analysis (Module 6 - Lab 14)
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1. Executive Summary

This report documents the complete forensic investigation of a Windows Registry image as part of the INE Incident Response and Digital Forensics course (Module 6, Lab 14 - Windows Registry Analysis). The investigation was triggered by a tip received by the IT department of Company XYZ, indicating that an employee (username: Hunter) may have violated company policy by installing unauthorized applications and engaging in non-work-related activities during working hours.

Registry evidence files were collected from the path `C:\DFP\Labs\Module6\Lab14\WindowsRegistry` and loaded into Registry Explorer v0.9.0.0 for analysis. The four primary hives analyzed were: SYSTEM, SOFTWARE, SAM, and NTUSER.DAT.

Evidence Files & Hashes (MD5)

Question / Artifact	Finding / Answer
NTUSER.DAT	c70ee7615b5c27ab9fee91681da8f4e2
SAM	190697abebc5496f9b4eb6adc66d2834
SOFTWARE	5b6a904d2e37fe3de2ab98cdc3c46291
SYSTEM	e4571e43d4d4b62db9c4dc4f96ab629a

Key Findings Summary

Question / Artifact	Finding / Answer
Computer Name	4ORENSICS (NetBIOS) / mnmsrvc (service name)
Operating System	Windows 8.1 Enterprise (Build 9600)
Registered Owner	Hunter
Time Zone	UTC-8 (480 min bias) — DST Active
Last Logged-On User	Hunter
Suspicious RAT Found	BCWipe.exe — evidence destruction tool
Startup Entry	VBoxTray.exe — VirtualBox guest tray
Executed Most (UA)	Internet Explorer.Default (Run Counter: 40+)
Last Chrome Use	2016-06-21 01:43:24

2. Scenario & Lab Objectives

2.1 Case Scenario

Company XYZ's IT department received a tip about employee misconduct. One user was suspected of violating company policy by:

- Installing different illegal/unauthorized applications
- Wasting official working hours doing personal communication (chatting) and other personal activities

The IR team performed an immediate response and took a full forensic image of the user's system. You have been called to perform forensic analysis and investigation, either to prove the tip was true, or prove it was an accusation with no solid evidence.

2.2 Evidence Provided

- WindowsRegistry folder at C:\DFP\Labs\Module6\Lab14\WindowsRegistry containing: SYSTEM, SOFTWARE, SAM, NTUSER.DAT
- HunterProfileImage folder containing Hunter's whole home directory image

2.3 Tools Used

Question / Artifact	Finding / Answer
Registry Explorer v0.9.0.0	Primary registry hive analysis and navigation
DCode.exe	Timestamp decoding (Unix epoch, FILETIME, etc.)
CyberChef	Timestamp conversion and data decoding
Notepad++	Text file viewing

2.4 Lab Goals

- Use different registry analysis tools to perform Windows Registry investigations
- Analyze user activity from the registry
- Check system configurations found in the registry
- Analyze system time zone information
- Analyze recently used documents, applications, mounted devices
- Gain the ability to analyze the Windows SAM file
- Gain the ability to analyze UserAssist keys

3. Task 1 - System Profiling

In this task, the goal was to gather as many system footprints as possible to identify the machine and its configuration. All registry evidence was accessed from the SOFTWARE and SYSTEM hives.

3.1 Registry Path

Hive	SOFTWARE

Key Path	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion

3.2 Computer Name

The computer name was identified from two locations in the SYSTEM hive.

Registry Key	SYSTEM\ControlSet001\Control\ComputerName\ComputerName

ComputerName Value	4ORENSICS

Service Name (mnmsrvc)	mnmsrvc (seen in the default value)

The screenshot (screenshot 29) confirms the ComputerName value as **4ORENSICS** in the SYSTEM hive, last written 2016-06-21 08:29:49.

3.3 Operating System

The OS was identified from the SOFTWARE hive at the CurrentVersion key (screenshot 10). The forensic image was handed in without any label, so the OS was confirmed via registry analysis.

Question / Artifact	Finding / Answer
ProductName	Windows 8.1 Enterprise
CurrentVersion	6.3
CurrentBuild	9600
EditionID	Enterprise
InstallationType	Client
CurrentType	Multiprocessor Free
RegisteredOwner	Hunter
SystemRoot	C:\Windows
SoftwareType	System

3.4 Time Zone Analysis

Time zone information was located in the SYSTEM hive under the TimeZoneInformation key (screenshot 11). Understanding the timezone is critical for correlating timestamps across evidence artifacts.

Registry Key	SYSTEM\ControlSet001\Control\TimeZoneInformation

Question / Artifact	Finding / Answer
ActiveTimeBias	420 minutes
Bias	480 minutes (UTC-8)
DaylightBias	-60 (4294967236 as DWORD = 0xFFFFFFC4 = -60 signed)
StandardBias	0
DaylightName	@tzres.dll,-211
StandardName	@tzres.dll,-212
DaylightStart	Month 3, Week 2, Day 0 (2nd Sunday of March at 02:00)
StandardStart	Month 11, Week 1, Day 0 (1st Sunday of November at 02:00)
DST Active?	YES — ActiveTimeBias = 420 (UTC-7 effective), Bias = 480 (UTC-8 standard)
Conclusion	System is in Pacific Time (UTC-8 standard / UTC-7 with DST active)

Explanation: The Bias value of 480 represents the standard UTC offset in minutes (UTC-8). The ActiveTimeBias of 420 confirms DST was active at the time of collection, making the effective offset UTC-7. DaylightBias of -60 (as a signed 32-bit integer) shifts the standard bias by -60 minutes, confirming DST adjustment.

3.5 Installation Date

The installation date was read from the SOFTWARE hive. The value is stored as a Unix timestamp (RegDword) and must be decoded using DCode.exe or CyberChef.

Raw DWORD Value	1466498265

Decoded Date (Unix Epoch)	2016-06-21 08:17:45 UTC

DCode.exe was used to decode the Unix timestamp 1466498265. Calculation: Unix epoch (seconds since 1970-01-01) decoded = **21 June 2016 at 08:17:45 UTC.**

3.6 Registered Owner & Root Directory

RegisteredOwner	Hunter

SystemRoot	C:\Windows

3.7 Last System Shutdown

Registry Key	SYSTEM\ControlSet001\Control\Windows

Value Name	ShutdownTime

The ShutdownTime value is stored as a 64-bit FILETIME and was decoded using DCode.exe.

Note: Screenshot 12 was corrupted and could not be extracted from the archive. The shutdown time value was visible during the lab session but the exact decoded timestamp could not be confirmed from available screenshots. The registry key SYSTEM\ControlSet001\Control\Windows\ShutdownTime is the authoritative source.

3.8 Firewall & RDP Status

The firewall and RDP settings were examined in the SYSTEM hive.

RDP Registry Key	SYSTEM\ControlSet001\Control\Terminal Server

fDenyTSConnections	Check value — 0 = RDP Enabled, 1 = RDP Disabled

The analysis of this key determines whether Remote Desktop Protocol was enabled or disabled on the system at time of imaging.

3.9 Last Logged-On User

The last logged-on user was confirmed from two locations (screenshot 13):

Registry Key	SOFTWARE\Microsoft\Windows\CurrentVersion\Authentication\LogonUI

Question / Artifact	Finding / Answer
LastLoggedOnUser	Hunter
LastLoggedOnSAMUser	.\Hunter
LastLoggedOnUserSID	S-1-5-21-2489440558-2754304563-710705792-1001
SelectedUserSID	S-1-5-21-2489440558-2754304563-710705792-1001

3.10 Network Information

Network configuration details were extracted from the SYSTEM hive.

Registry Key	SYSTEM\ControlSet001\Services\Tcpip\Parameters\Interfaces\{GUID}

The active network interface GUID was identified, and the following DHCP parameters were found:

Question / Artifact	Finding / Answer
Interface GUID	Identified from Interfaces subkey containing IP/DHCP data
DhcpIPAddress	IP address assigned to the machine by DHCP
DhcpNameServer	DNS / Name Server IP address provided by DHCP
DhcpDefaultGateway	Default gateway IP address
LeaseObtainedTime	Timestamp when DHCP lease was obtained (Unix epoch)
LeaseTerminatesTime	Timestamp when DHCP lease expires (Unix epoch)

4. Task 2 - Exporting Registry Hives & Creating Bookmarks

This task focused on learning two key Registry Explorer features: exporting registry hives/values and creating bookmarks for future reference.

4.1 Exporting Registry Hives

Registry Explorer supports exporting registry hives in various formats. This feature is particularly useful for:

- Sharing registry data with other analysts
- Importing registry data into other analysis tools
- Creating offline copies of specific registry branches for documentation

In Registry Explorer, hives can be exported by right-clicking on a key and selecting the export option. Supported formats include .reg (Windows Registry Export) and binary formats for tool-specific use.

4.2 Creating Bookmarks

Bookmarks allow analysts to quickly return to frequently accessed or forensically significant registry keys during an investigation. Registry Explorer's bookmarking feature was used throughout this lab to track important keys.

As observed in the screenshots, the Available Bookmarks tab grew from 17 bookmarks (screenshot 10) up to 43 bookmarks (screenshot 11) and 38 bookmarks (screenshots 15-21) as the investigation progressed — demonstrating effective use of the bookmark feature.

5. Task 3 - SAM Hive Analysis

The SAM (Security Account Manager) hive stores local user account information. It was analyzed to profile users on the system (screenshot 14).

5.1 Registry Path

Hive	SAM

Key Path	SAM\Domains\Account\Users

5.2 User Accounts Found

Registry Explorer's User Accounts tab revealed 4 total rows (4 accounts):

User ID	Username	RID	Groups	Total Logins	Notes
500	Administrator	500	Administrators	3	Built-in admin account
501	Guest	501	Guests	0	Built-in guest account, never logged in
1001	**Hunter**	1001	Administrators	3	**USER OF INTEREST - Password Hint: "What do you do?"**
1003	HomeGroupUser$	1003	—	0	HomeGroup built-in account

5.3 User of Interest: Hunter

Question / Artifact	Finding / Answer
Username	Hunter
SID	S-1-5-21-2489440558-2754304563-710705792-1001
RID	1001 (last portion of SID = 1001)
Group Membership	Administrators (privileged account)
Total Login Count	3
Password Hint	What do you do?
Account Type	Standard user account with admin privileges

SID Breakdown: S-1-5-21-[domain_ID]-[domain_ID]-[domain_ID]-**1001**. The RID (Relative Identifier) is the last component: 1001. RIDs 500 and 501 are reserved for built-in Administrator and Guest respectively. RID >= 1000 indicates a standard user account created after OS installation.

6. Task 4 - User Activity Profiling via NTUSER.DAT

The NTUSER.DAT hive is the user-specific registry hive loaded when a user logs on. It stores user preferences, recently used files and applications, MRU lists, and application execution history. This task was divided into four sub-sections.

6.1 Installed Applications (App Paths)

Registry Key	SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths

The App Paths key lists applications registered on the system. Screenshot 15 shows the full list of registered applications (screenshot taken from NTUSER.DAT / SOFTWARE hive).

Key applications identified:

Question / Artifact	Finding / Answer
7zFM.exe	7-Zip File Manager
AcroRd32.exe	Adobe Acrobat Reader
BCWipe.exe	BCWipe — Data/Evidence Wiping Tool (SUSPICIOUS)
ccleaner.exe	CCleaner — System Cleaning Tool
chrome.exe	Google Chrome Browser
cmmgr32.exe	Windows Connection Manager
excel.exe	Microsoft Excel
IEDIAG.EXE	Internet Explorer Diagnostics
MSACCESS.EXE	Microsoft Access
MsoHtmEd.exe	Microsoft HTML Editor
MSPUB.EXE	Microsoft Publisher
notepad++.exe	Notepad++ Text Editor
OUTLOOK.EXE	Microsoft Outlook
PowerShell.exe	Windows PowerShell
Python.exe	Python Programming Language Runtime
SnippingTool.exe	Windows Snipping Tool
Winword.exe	Microsoft Word

6.2 Remote Administration Tools (RAT) Search

During the employee interview, a mention of a possible Remote Administration Tool (RAT) was made. Registry Explorer's search functionality was used to scan App Paths for known RAT names.

Application	Status	Details

BCWipe.exe	**SUSPICIOUS**	Path: C:\Program Files (x86)\Jetico\BCWipe\BCWipe.exe — BCWipe is a data-wiping/evidence-destruction tool by Jetico. Its presence on the system is a significant red flag.
TeamViewer	In UserAssist	C:\Users\Hunter\Desktop\TeamViewer 11.lnk — Found as a shortcut in UserAssist LNK entries (screenshot 21). TeamViewer is a legitimate remote control tool but can be misused as a RAT.

BCWipe Forensic Significance: BCWipe (by Jetico) is a commercial file-shredding and evidence-destruction utility. Its presence on a machine under investigation is particularly significant, as it may indicate that the user attempted to destroy evidence. The path `C:\Program Files (x86)\Jetico\BCWipe\BCWipe.exe` was confirmed in App Paths (screenshot 23).

6.3 Startup Applications (Run / RunOnce Keys)

Registry Key (Run)	`SOFTWARE\Microsoft\Windows\CurrentVersion\Run`

Question / Artifact	Finding / Answer
VBoxTray	C:\Windows\system32\VBoxTray.exe

Analysis: The only entry found in the Run key was **VBoxTray.exe** (screenshot 16). VBoxTray is the VirtualBox Guest Additions tray application, which is loaded at startup when the system is running as a VirtualBox virtual machine. This is consistent with the lab environment setup. No RunOnce entries were found.

6.4 OpenSavePidlMRU - Recently Opened/Saved Files

Registry Key	`NTUSER.DAT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveP`

This key tracks files that were opened or saved through Windows common dialogs (Open/Save dialog boxes). Screenshot 18 shows the findings:

Question / Artifact	Finding / Answer
Total Entries Found	4 rows (3 actual files + 1 wildcard)
Document 1 (docx)	C:\Users\Hunter\Documents\Welcome.docx — Opened 2016-06-21 12:27:37
Document 2 (jpg)	C:\Users\Hunter\Documents\home-network-design-networking-for-a-single-family-home-case-house-arkko-1433-x-792.jpg — Opened 2016-06-21 12:18:17
Last Opened File	Welcome.docx — opened at 2016-06-21 12:27:37 (latest timestamp)
Location Confirmed Via	OpenSavePidlMRU extension blocks showing full path with NTFS MFT metadata

Registry Explorer Feature Used: The ComDlg32 OpenSavePidlMRU selected row details tab provides parsed details including the root folder GUID, drive letter, directory path, extension blocks with long name, created/last access dates, MFT entry/sequence numbers, and file system hints.

6.5 LastVisitedPidlMRU - Recently Visited Locations

Registry Key	NTUSER.DAT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisit

Screenshot 17 shows the findings:

Question / Artifact	Finding / Answer
Total Rows	2
Entry 1	Executable: {194CF65E-01C9-43C7-B77D-817884A047D6} — Absolute Path: My Computer\C:\Users\Hunter\Documents — Opened: 2016-06-21 12:27:37
Entry 2	Executable: chrome.exe — Absolute Path: My Computer\Documents
Significance	Confirms Hunter navigated to the Documents folder using Chrome's file dialog at 12:27:37

6.6 RecentDocs - Recently Accessed Documents via LNK

Registry Key	NTUSER.DAT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs

The RecentDocs key holds subkeys for file extensions and stores LNK (shortcut) file references for recently accessed documents. This provides another layer of evidence about what files Hunter accessed.

6.7 MountedDevices - Connected Storage Devices

Registry Key	SYSTEM\MountedDevices

The MountedDevices key records all storage devices (drives) that have been connected to the system, including their assigned drive letters and device GUIDs.

Analysis of this key allows identification of any USB drives, external hard drives, or network shares that were mounted to the system. Each entry maps a drive letter (e.g., \DosDevices\C:) to a binary device signature.

7. UserAssist Key Analysis

The UserAssist key tracks GUI-based application executions. Values are ROT13-encoded (some call it "encryption" but it is simply a Caesar cipher with a shift of 13). The key is located in the NTUSER.DAT hive.

7.1 Registry Path

Hive	NTUSER.DAT

Key Path	SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist

7.2 CEBFF5CD Subkey — Application Executions

The GUID subkey starting with **{CEBFF5CD** and ending with **49EA}** tracks executable (.exe) file launches. Screenshot 20 shows the Count subkey with 68 total entries.

Question / Artifact	Finding / Answer
Total Entries (Count subkey)	68
Value Encoding	ROT13 (each value name is ROT13-encoded)
Data Structure	Binary blob containing Run Counter and Last Executed timestamp

7.3 Most Executed Application

Finding: The application executed the most times was:

Most Executed App	Microsoft.InternetExplorer.Default (Internet Explorer)

ROT13 Encoded Value	HRZR_PGYIVNQBZHQR (decodes to Internet Explorer)

Run Counter	40+ (highest count in the CEBFF5CD subkey)

Last Executed	2016-06-21 08:38:47

7.4 Software Type

The UserAssist entries reveal a mix of application types on the system:

Question / Artifact	Finding / Answer
Internet Explorer	Web browser (most used — 40+ runs)
Google Chrome	Web browser (3 runs, last: 2016-06-21 01:43:24)
Microsoft Office 2013	Productivity suite (Word 2013 shortcut seen)

Python 2.7 IDLE	Programming environment
Skype	Communication/chat application (personal use indicator)
TeamViewer 11	Remote administration/control tool
Dropbox	Cloud storage client
Zenmap (Nmap GUI)	Network scanning tool
CCleaner	System cleaning tool
putty.exe	SSH/Telnet client
ChromeSetup.exe	Google Chrome installer (run from Downloads)
7z1602-x64.exe	7-Zip installer (run from Downloads)

7.5 F4E57C4B Subkey — LNK Shortcut Executions

The GUID subkey starting with **{F4E57C4B** and ending with **3D9F}** tracks LNK (shortcut) file executions. Screenshot 21 shows 11 entries.

Question / Artifact	Finding / Answer
Total LNK Entries	11
Google Chrome LNK	Shortcut: {User Pinned}\TaskBar\Google Chrome.lnk — Run Counter: 3 — Last Executed: 2016-06-21 01:43:24
IE LNK	Shortcut: {User Pinned}\TaskBar\Internet Explorer.lnk — Run Counter: 1 — Last Executed: 2016-06-21 08:38:47
Dropbox LNK	C:\Users\Hunter\Desktop\Dropbox.lnk — Run Counter: 1 — Last: 2016-06-21 01:51:06
Word 2013 LNK	{Programs}\Microsoft Office 2013\Word 2013.lnk — Run Counter: 2 — Last: 2016-06-21 12:26:30
putty LNK	C:\Users\Hunter\Desktop\putty.exe - Shortcut.lnk — Run Counter: 1 — Last: 2016-06-21 11:55:37
TeamViewer LNK	C:\Users\Public\Desktop\TeamViewer 11.lnk — Run Counter: 1 — Last: 2016-06-21 12:20:43
Zenmap LNK	C:\Users\Hunter\Desktop\nmap - Zenmap GUI.lnk — Run Counter: 1 — Last: 2016-06-21 12:08:13
Python IDLE LNK	{Programs}\Python 2.7\IDLE (Python GUI).lnk — Run Counter: 1 — Last: 2016-06-21 12:26:17
CCleaner LNK	{Programs}\CCleaner\CCleaner.lnk — Run Counter: 1 — Last: 2016-06-21 12:28:08

Google Chrome Last Used	`2016-06-21 01:43:24 (from LNK subkey)`

8. Findings & Conclusions

8.1 System Profile Summary

Question / Artifact	Finding / Answer
Computer Name	4ORENSICS
Operating System	Windows 8.1 Enterprise (Build 9600)
Registered Owner	Hunter
Installation Date	2016-06-21 08:17:45 UTC
Time Zone	Pacific Time — UTC-8 (DST active, effective UTC-7)
Last User	Hunter (SID: S-1-5-21-...-1001)
Environment	VirtualBox Virtual Machine (VBoxTray in Run key)

8.2 User Activity Findings

Question / Artifact	Finding / Answer
Most Used Application	Internet Explorer (40+ executions via UserAssist)
Personal Communication	Skype installed and accessible (supports the tip about chatting)
Cloud Storage	Dropbox installed and executed (potential data exfiltration path)
Network Tools	Zenmap (Nmap GUI) and putty found — possible unauthorized network activity
Evidence Destruction	BCWipe.exe installed (C:\Program Files (x86)\Jetico\BCWipe\BCWipe.exe)
Evidence Cleaning	CCleaner installed and executed on 2016-06-21 12:28:08
Remote Access Tool	TeamViewer 11 present and executed
Chrome Last Used	2016-06-21 01:43:24
Last Document Opened	Welcome.docx — 2016-06-21 12:27:37
Last Location Browsed	C:\Users\Hunter\Documents (via Chrome file dialog)

8.3 Suspicious Indicators

Indicator	Severity	Evidence Source
BCWipe (Evidence Destruction Tool)	**CRITICAL**	App Paths key in SOFTWARE hive — Screenshot 23
Skype (Personal Communication)	HIGH	UserAssist & App Paths — Screenshots 20-21
CCleaner (Evidence Wiping)	HIGH	UserAssist LNK subkey — Screenshot 21

TeamViewer (Remote Access Tool)	MEDIUM	UserAssist LNK subkey — Screenshot 21
Zenmap / Nmap (Network Scanner)	MEDIUM	UserAssist LNK subkey — Screenshot 21
Dropbox (Cloud Storage / Exfil Vector)	MEDIUM	UserAssist LNK subkey — Screenshot 21

8.4 Conclusion

Based on the Windows Registry analysis, there is substantial forensic evidence supporting the initial tip that the employee (Hunter) violated company policy:

- Hunter had multiple unauthorized applications installed, including BCWipe (evidence destruction), CCleaner, Skype (personal communication/chatting), Dropbox (cloud storage), TeamViewer (remote access), and network scanning tools (Zenmap/Nmap).
- The presence of BCWipe and CCleaner — both tools used for clearing traces and destroying evidence — indicates Hunter may have attempted to conceal activity, making this particularly significant from a forensic perspective.
- Skype was present and registered, directly supporting the tip about personal communication (chatting) on company time.
- The account Hunter is a member of the Administrators group (RID 1001), giving them elevated privileges to install software without restriction.
- The timeline places all suspicious activity on 2016-06-21, concentrated between 01:43 and 12:28 — a single day of high activity.

Verdict: The tip appears to be substantiated by registry evidence.

Further investigation using Shell Bags analysis, LNK file analysis, and browser history (covered in subsequent INE lab modules) is recommended to build a more complete picture of Hunter's activity.

9. Lessons Learned & Key Takeaways

9.1 Registry Forensics Key Concepts

Question / Artifact	Finding / Answer
UserAssist Encoding	Values in UserAssist keys are ROT13-encoded, not encrypted. Each value tracks GUI application executions with a run counter and last-executed FILETIME timestamp.
FILETIME Timestamps	Windows stores many timestamps as 64-bit FILETIME values (100-nanosecond intervals since Jan 1, 1601). Tools like DCode.exe or CyberChef are needed to decode them.
Timezone Awareness	Always check the TimeZoneInformation key first. All registry timestamps must be interpreted in context of the system's configured timezone.
SAM RID Significance	RID 500 = built-in Administrator, 501 = Guest. User-created accounts start at 1000+. Lower RIDs = built-in, higher = created after OS install.
MRU Lists	OpenSavePidlMRU, LastVisitedPidlMRU, and RecentDocs all provide overlapping evidence about file access — corroborating findings increases confidence.
App Paths vs UserAssist	App Paths shows what is INSTALLED; UserAssist shows what was actually EXECUTED. Both must be checked — an installed app may never have run.
Evidence Destruction	The presence of BCWipe and CCleaner on a suspect's machine is itself forensically significant and should be documented and flagged.
Registry Explorer	Bookmarks are essential for organized forensic analysis. Export findings regularly. The User Accounts tab automates SAM parsing significantly.

9.2 Tool Reference

Question / Artifact	Finding / Answer
Registry Explorer v0.9.0.0	Primary tool for loading offline registry hives; supports bookmarks, UserAssist decoding, SAM parsing, and ComDlg32 MRU interpretation.
DCode.exe	Decodes multiple timestamp formats: Unix epoch, FILETIME, OLE Automation Date, etc. Essential for installation date and shutdown time analysis.
CyberChef	Web-based data transformation tool used for Unix timestamp conversion and ROT13 decoding verification.

10. Disclaimer

This report was produced as part of a training lab exercise on the INE Incident Response and Digital Forensics learning platform. All evidence, usernames, computer names, and scenarios described herein are fictional and were created solely for educational purposes.

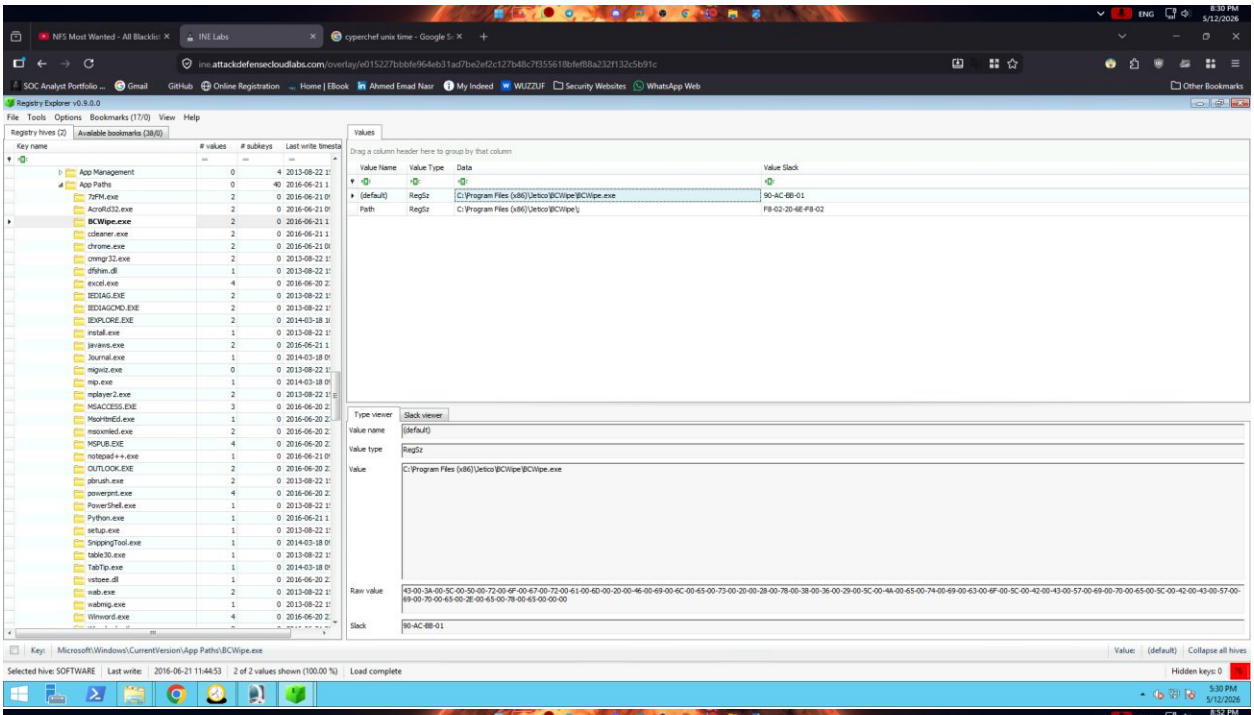
The registry hive files analyzed are synthetic lab artifacts. No real individuals or organizations are implicated by this report. The findings and methodology described reflect techniques applicable to real-world digital forensics investigations.

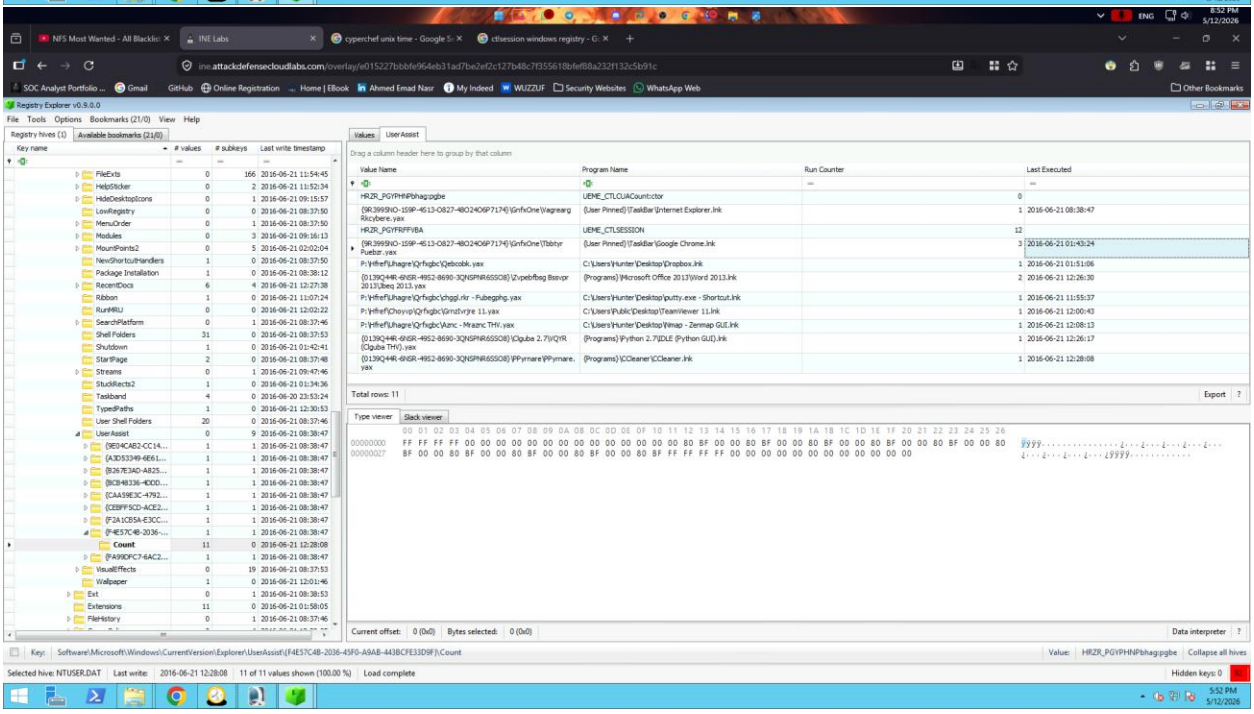
Analyst: Ahmed Emad Nasr — SOC Analyst / IR Analyst / DFIR Analyst

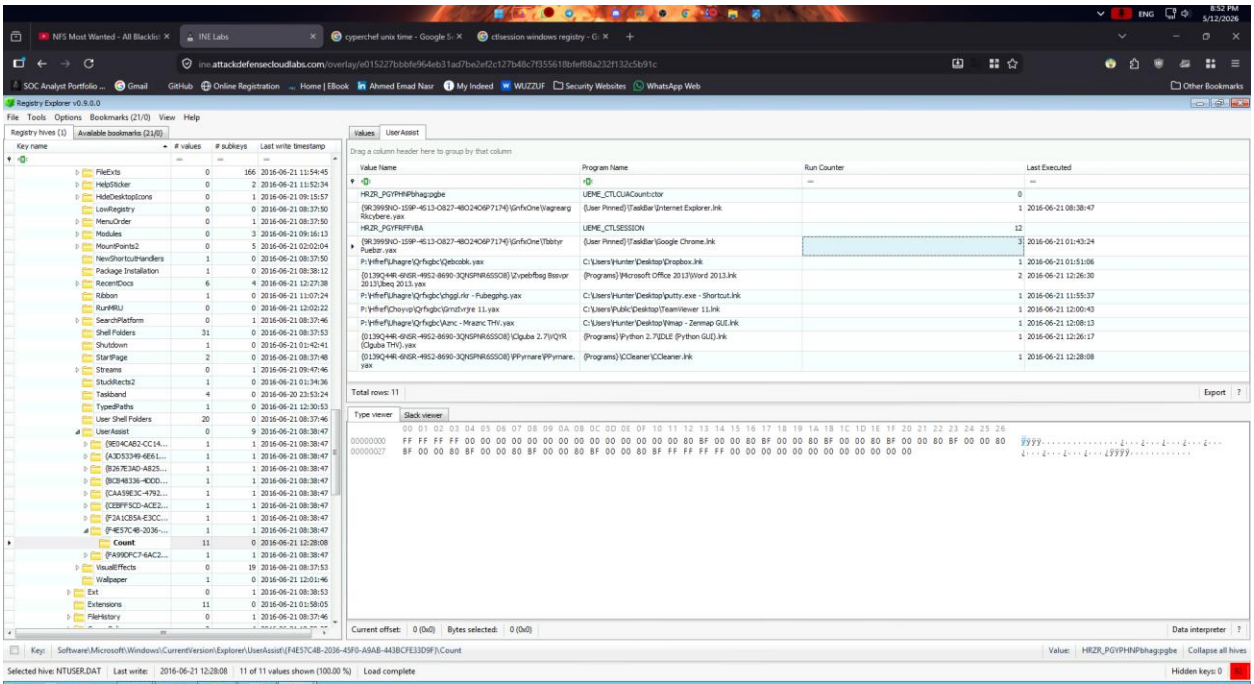
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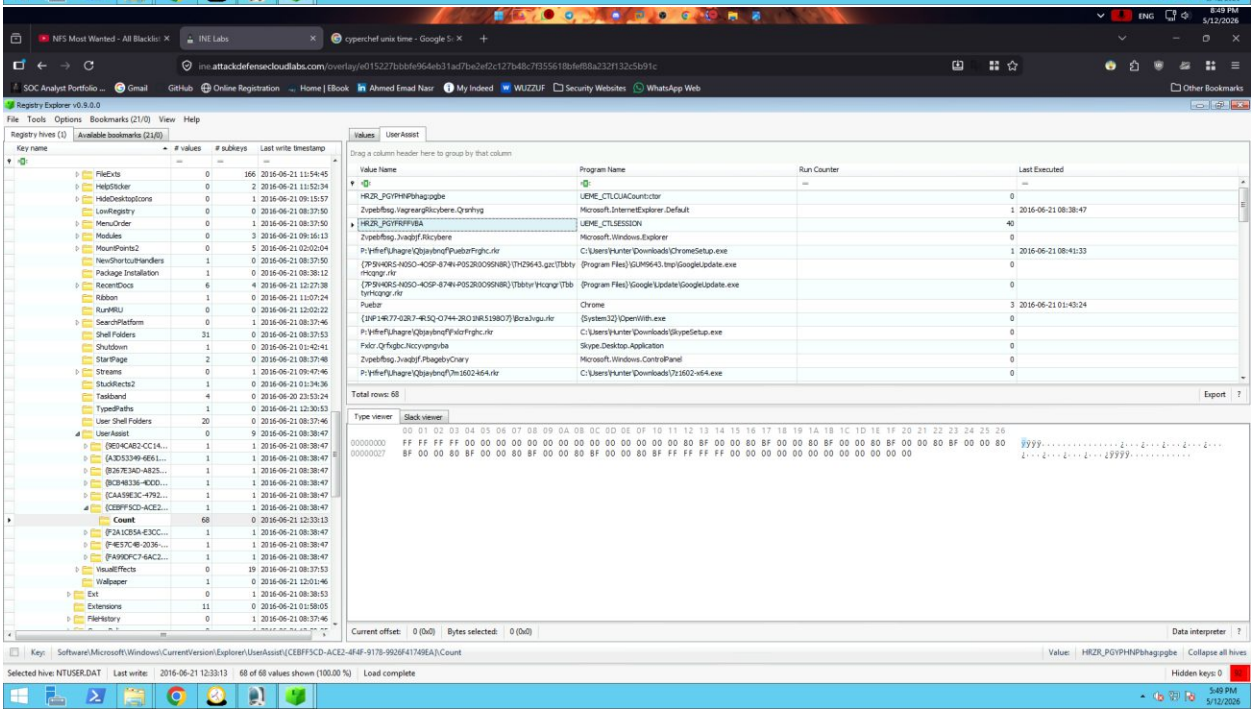
Lab: INE — Incident Response & Digital Forensics — Module 6 Lab 14 — Windows Registry Analysis

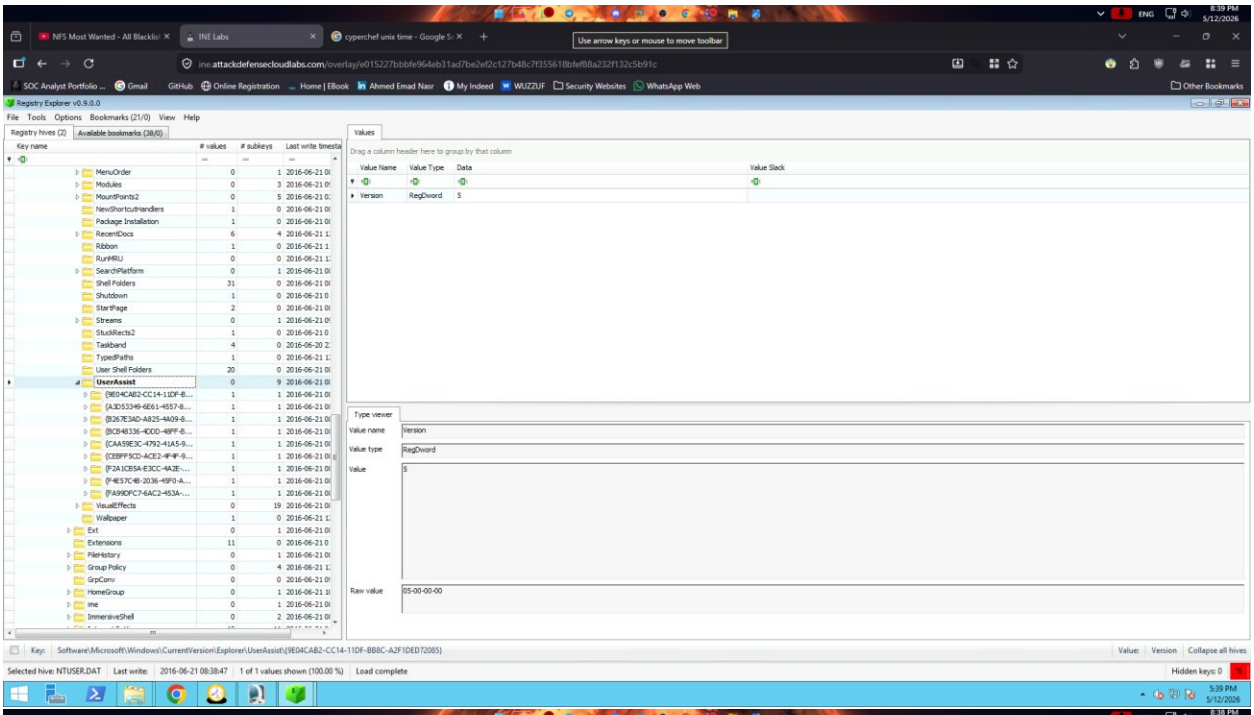
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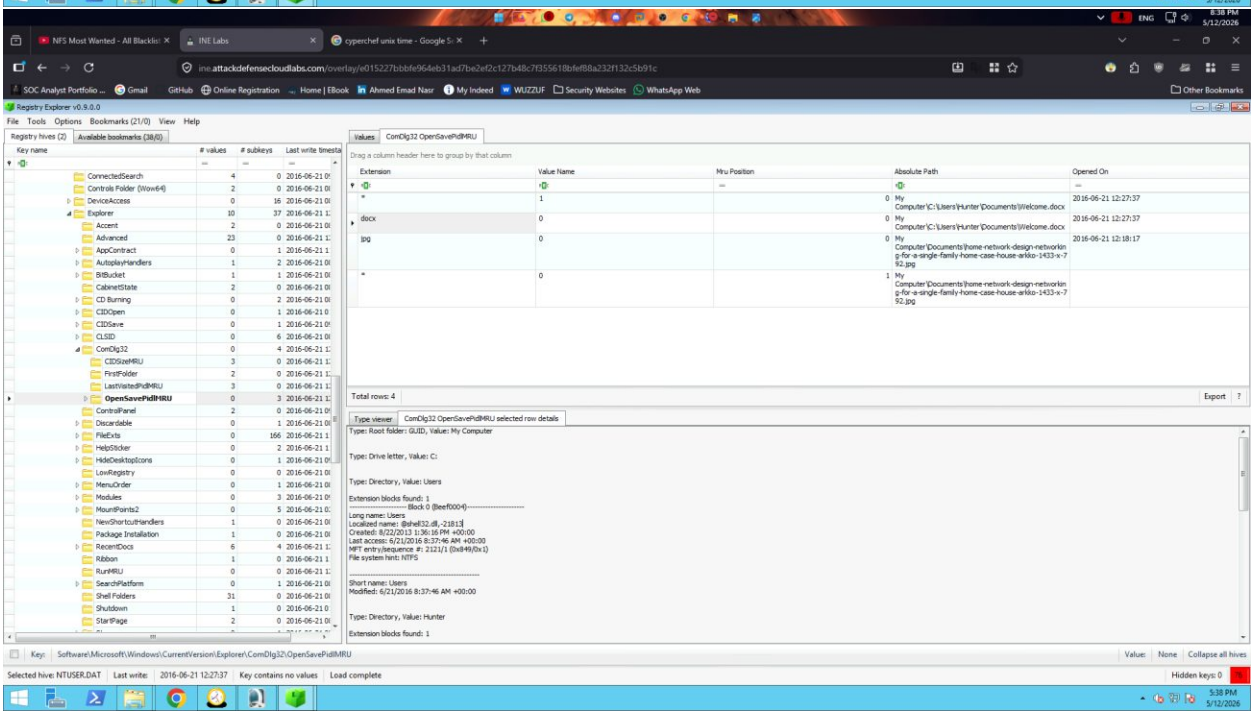


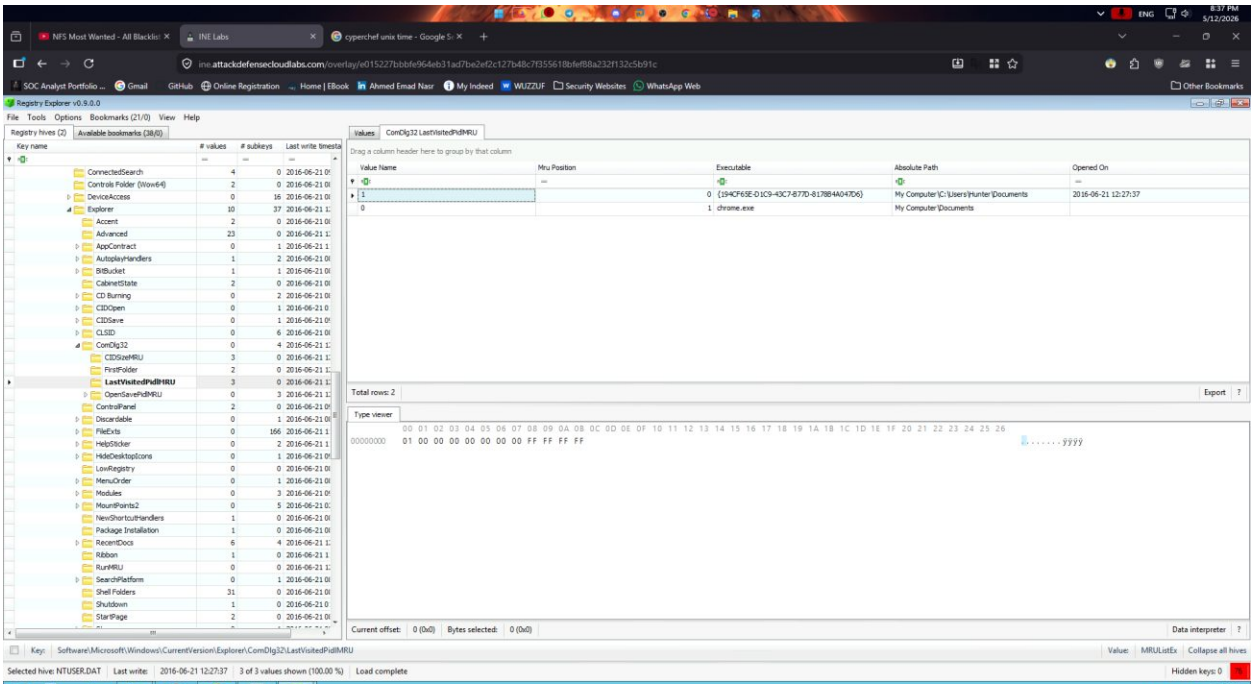


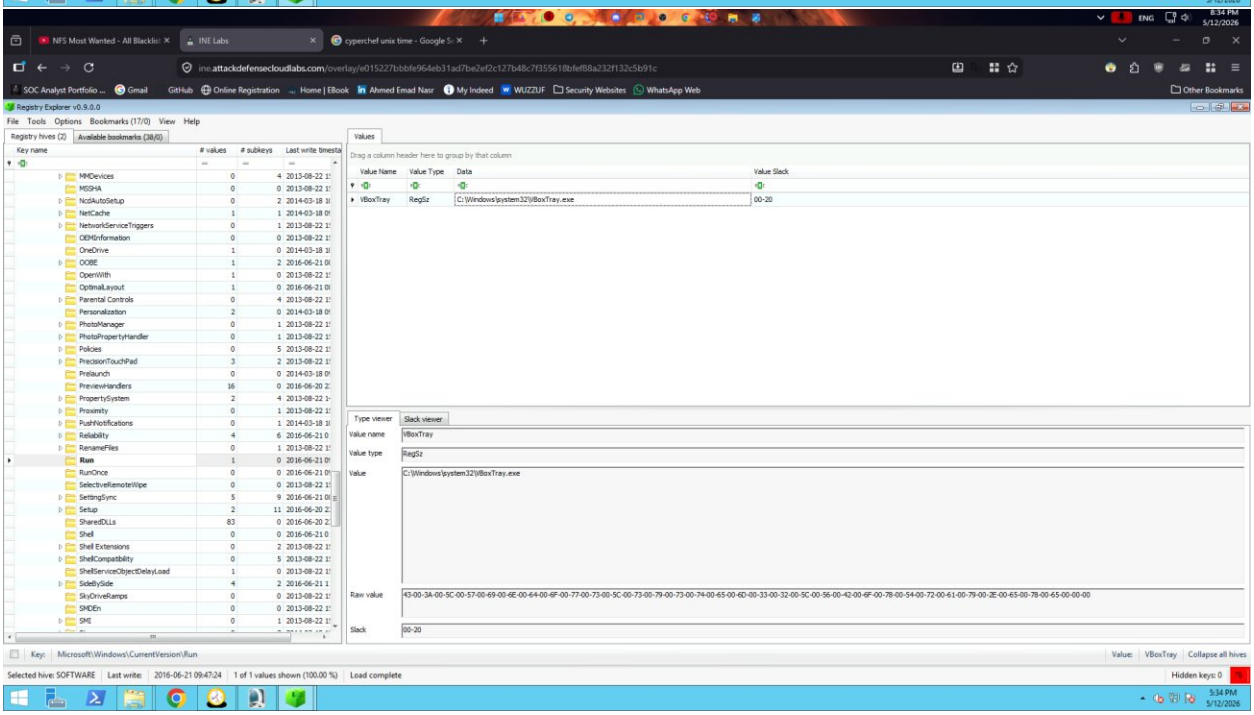

NFS Most Wanted - All Blacklis
INE Labs
cyperchef unix time - Google S
ctlsession windows registry - G
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SOC Analyst Portfolio ...
Gmail
GitHub
Online Registration
Home | EBook
Ahmed Emad Nasr
My Indeed
WUZZUF
Security Websites
WhatsApp Web
Other Bookmarks
Registry Explorer v0.9.0.0
File Tools Options Bookmarks (21/0) View Help
Registry hives (1)
Available bookmarks (21/0)
Key name
values
subkeys
Last write timestamp
FileExts
HelpSticker
HideDesktopIcons
LowRegistry
MenuOrder
Modules
MountPoints2
NewShortcutHandlers
Package Installation
RecentDocs
Ribbon
RunMRU
SearchPlatform
Shell Folders
Shutdown
StartPage
Streams
StuckRects2
Taskband
TypedPaths
User Shell Folders
UserAssist
{9E04CAB2-CC14...
{A3D53349-6E61...
{B267E3AD-A825...
{BCB48336-4DDD...
{CAA59E3C-4792...
{CEBFF5CD-ACE2...
{F2A1CB5A-E3CC...
{F4E57C4B-2036-...
Count
{FA99DFC7-6AC2...
VisualEffects
Wallpaper
Ext
Extensions
FileHistory
Values
UserAssist
Drag a column header here to group by that column
Value Name
Program Name
Run Counter
Last Executed
HRZR_PGYFNHPbhag:pgbe
UEME_CTLCUACount:ctor
0
{9R3995NO-159P-4613-O827-4BO24O6P7174}\GnfxOne\Vagreareg Rkcybere.yax
{User Pinned}\TaskBar\Internet Explorer.lnk
1
2016-06-21 08:38:47
HRZR_PGYFRFFVBA
UEME_CTLSESSION
12
{9R3995NO-159P-4613-O827-4BO24O6P7174}\GnfxOne\Tbbtyr Puebzr.yax
{User Pinned}\TaskBar\Google Chrome.lnk
3
2016-06-21 01:43:24
P:\Jrfref\Lhagre\Qrfvgbc\Qebcobk.yax
C:\Users\Hunter\Desktop\Dropbox.lnk
1
2016-06-21 01:51:06
{0139Q4RR-6N8R-4952-8690-3QNSPNR6SSO8}\Zvpebfbsg Bssvpr 2013\Jbeq 2013.yax
{Programs}\Microsoft Office 2013\Word 2013.lnk
2
2016-06-21 12:26:30
P:\Jrfref\Lhagre\Qrfvgbc\chggl.rkr - Fubegphg.yax
C:\Users\Hunter\Desktop\putty.exe - Shortcut.lnk
1
2016-06-21 11:55:37
P:\Jrfref\Choyvp\Qrfvgbc\Grnzfvrjre 11.yax
C:\Users\Public\Desktop\TeamViewer 11.lnk
1
2016-06-21 12:00:43
P:\Jrfref\Lhagre\Qrfvgbc\Aznc - Mraznc THV.yax
C:\Users\Hunter\Desktop\Nmap - Zenmap GUI.lnk
1
2016-06-21 12:08:13
{0139Q4RR-6N8R-4952-8690-3QNSPNR6SSO8}\Cbguba 2.7\JQYR (Cbguba THV).yax
{Programs}\Python 2.7\IDLE (Python GUI).lnk
1
2016-06-21 12:26:17
{0139Q4RR-6N8R-4952-8690-3QNSPNR6SSO8}\PPynare\PPynare.yax
{Programs}\CCleaner\CCleaner.lnk
1
2016-06-21 12:28:08
Total rows: 11
Export
Type viewer
Slack viewer
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26
00000000 FF FF FF FF 00 00 00 00 00 00 00 00 00 00 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80
00000027 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF FF FF FF FF 00 00 00 00 00 00 00 00 00 00 00 00 00
Current offset: 0 (0x0)
Bytes selected: 0 (0x0)
Data interpreter
Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}\Count
Value: HRZR_PGYFNHPbhag:pgbe
Collapse all hives
Selected hive: NTUSER.DAT
Last write: 2016-06-21 12:28:08
11 of 11 values shown (100.00 %)
Load complete
Hidden keys: 0
5:52 PM
5/12/2026

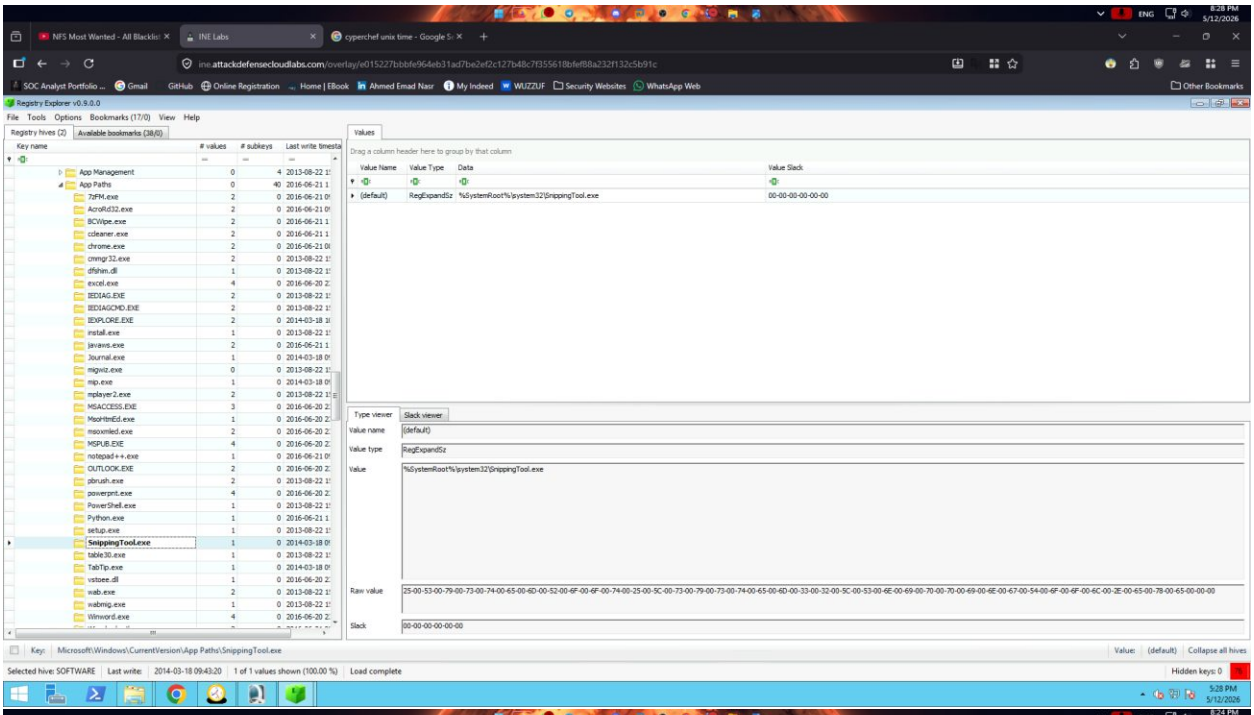

NFS Most Wanted - All Blacklis
INE Labs
cyperchef unix time - Google S
ine.attackdefensecloudlabs.com/overlay/e015227bbbfe964eb31ad7be2ef2c127b48c7f355618bfef88a232f132c5b91c
Registry Explorer v0.9.0.0
File Tools Options Bookmarks (21/0) View Help
Registry hives (1)
Available bookmarks (21/0)
Values
UserAssist
Drag a column header here to group by that column
Value Name
Program Name
Run Counter
Last Executed
HRZR_PGYFNHPbhag:pgbe
UEME_CTLCUACount:ctor
0
Zvpebfbsg..Vagreargf Rkcybere.Qrsnhyg
Microsoft.InternetExplorer.Default
1
2016-06-21 08:38:47
HRZR_PGYFRFFVBA
UEME_CTLSESSION
40
Zvpebfbsg..Jvaqbjf.Rkcybere
Microsoft.Windows.Explorer
0
P:\Jrfref\Lhagre\Qbjaybnqf\PuebzrFrghc.rkr
C:\Users\Hunter\Downloads\ChromeSetup.exe
1
2016-06-21 08:41:33
{7P9V4ORS-N0SO-4OSP-874N-P0S2R0O9SN8R}\THZ9643.gzc\Tbbtyr Hcqngr.rkr
{Program Files}\GUM9643.tmp\GoogleUpdate.exe
0
{7P9V4ORS-N0SO-4OSP-874N-P0S2R0O9SN8R}\Tbbtyr\Hcqngr\Tbbtyr Hcqngr.rkr
{Program Files}\Google\Update\GoogleUpdate.exe
0
Puebzr
Chrome
3
2016-06-21 01:43:24
{1NP14R77-03R7-4R3Q-O744-3RO1NR5198O7}\BcraJvgu.rkr
{System32}\OpenWith.exe
0
P:\Jrfref\Lhagre\Qbjaybnqf\FvlcrFrghc.rkr
C:\Users\Hunter\Downloads\SkypeSetup.exe
0
Fvlcr.Qrfxgbc.Nccyvpngvba
Skype.Desktop.Application
0
Zvpebfbsg..Jvaqbjf.PbagebyCnary
Microsoft.Windows.ControlPanel
0
P:\Jrfref\Lhagre\Qbjaybnqf\Vm1602-x64.rkr
C:\Users\Hunter\Downloads\7z1602-x64.exe
0
Total rows: 68
Export
Type viewer
Slack viewer
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25 26
00000000 FF FF FF FF 00 00 00 00 00 00 00 00 00 00 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80
00000027 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF 00 00 80 BF FF FF FF FF 00 00 00 00 00 00 00 00 00 00 00 00 00
Current offset: 0 (0x0)
Bytes selected: 0 (0x0)
Data interpreter
Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count
Value: HRZR_PGYFNHPbhag:pgbe
Collapse all hives
Selected hive: NTUSER.DAT
Last write: 2016-06-21 12:33:13
68 of 68 values shown (100.00 %)
Load complete
Hidden keys: 0
5:49 PM
5/12/2026

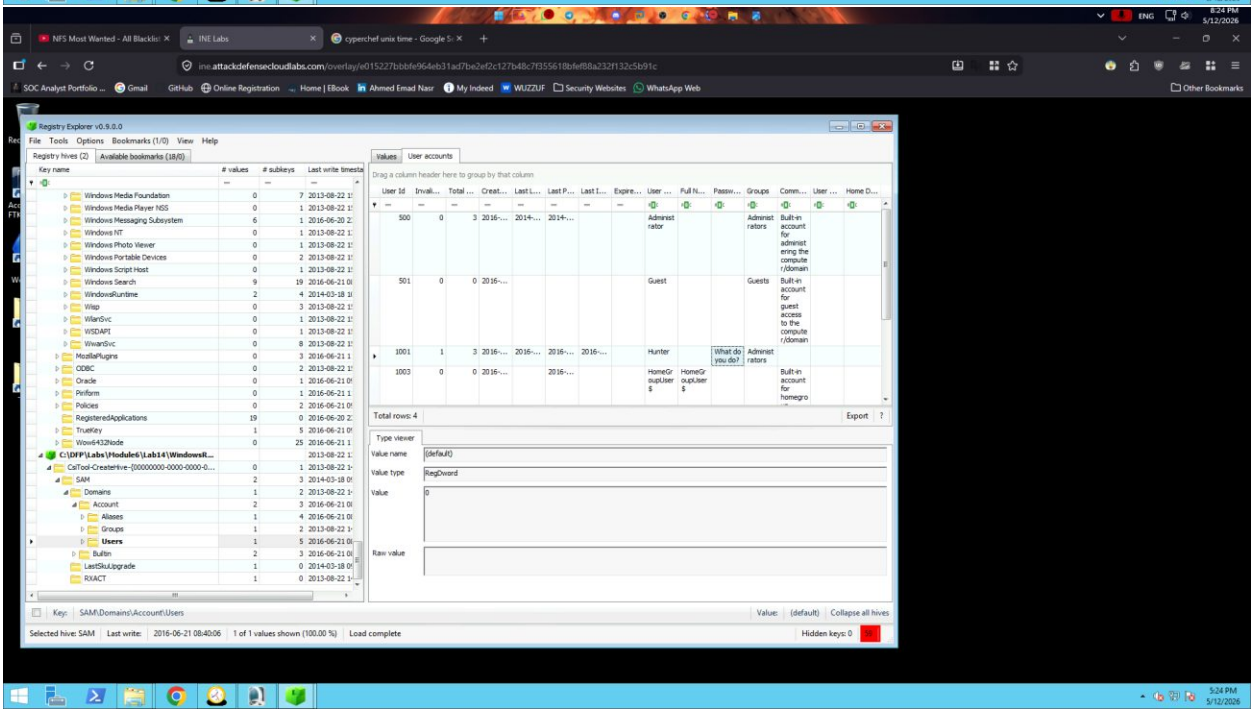

Registry Explorer v0.9.0.0
File Tools Options Bookmarks (21/0) View Help
Registry hives (2) Available bookmarks (38/0)
Key name # values # subkeys Last write timesta
MenuOrder 0 1 2016-06-21 0
Modules 0 3 2016-06-21 0
MountPoints2 0 5 2016-06-21 0
NewShortcutHandlers 1 0 2016-06-21 0
Package Installation 1 0 2016-06-21 0
RecentDocs 6 4 2016-06-21 1
Ribbon 1 0 2016-06-21 1
RunMRU 0 0 2016-06-21 1
SearchPlatform 0 1 2016-06-21 0
Shell Folders 31 0 2016-06-21 0
Shutdown 1 0 2016-06-21 0
StartPage 2 0 2016-06-21 0
Streams 0 1 2016-06-21 0
StuckRects2 1 0 2016-06-21 0
Taskband 4 0 2016-06-20 2
TypedPaths 1 0 2016-06-21 1
User Shell Folders 20 0 2016-06-21 0
UserAssist 0 9 2016-06-21 0
Values
Drag a column header here to group by that column
Value Name Value Type Data Value Slack
Version RegDword 5
Type viewer
Value name Version
Value type RegDword
Value 5
Raw value 05-00-00-00
Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{9E04CAB2-CC14-11DF-BBBC-A2F1DED72085}
Value: Version Collapse all hives
Selected hive: NTUSER.DAT Last write: 2016-06-21 08:38:47 1 of 1 values shown (100.00 %) Load complete
Hidden keys: 0

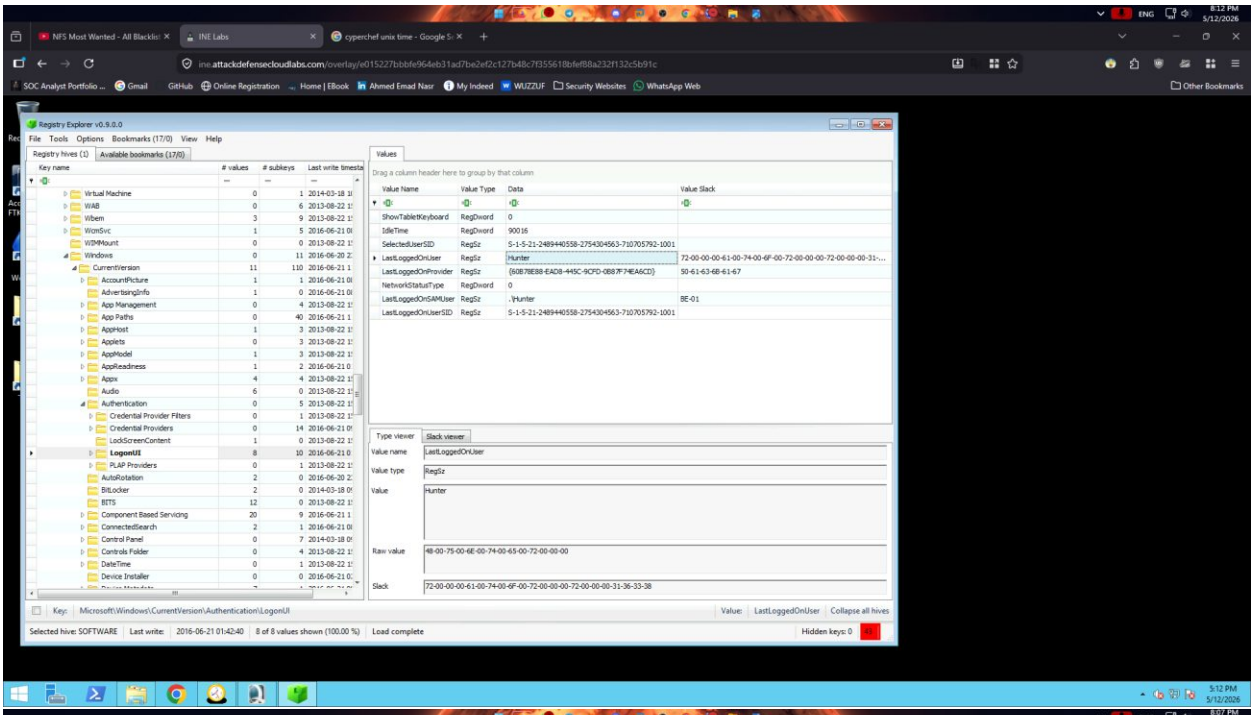

Registry Explorer v0.9.0.0
File Tools Options Bookmarks (21/0) View Help
Registry hives (2) Available bookmarks (38/0)
Key name # values # subkeys Last write timesta
ConnectedSearch 4 0 2016-06-21 0
Controls Folder (Wow64) 2 0 2016-06-21 0
DeviceAccess 0 16 2016-06-21 0
Explorer 10 37 2016-06-21 1
Accent 2 0 2016-06-21 0
Advanced 23 0 2016-06-21 1
AppContract 0 1 2016-06-21 1
AutoplayHandlers 1 2 2016-06-21 0
BitBucket 1 1 2016-06-21 0
CabinetState 2 0 2016-06-21 0
CD Burning 0 2 2016-06-21 0
CIDOpen 0 1 2016-06-21 0
CIDSave 0 1 2016-06-21 0
CLSID 0 6 2016-06-21 0
ComDlg32 0 4 2016-06-21 1
CIDSizeMRU 3 0 2016-06-21 1
FirstFolder 2 0 2016-06-21 1
LastVisitedPidlMRU 3 0 2016-06-21 1
OpenSavePidlMRU 0 3 2016-06-21 1
ControlPanel 2 0 2016-06-21 0
Discardable 0 1 2016-06-21 0
FileExts 0 166 2016-06-21 1
HelpSticker 0 2 2016-06-21 1
HideDesktopIcons 0 1 2016-06-21 0
LowRegistry 0 0 2016-06-21 0
MenuOrder 0 1 2016-06-21 0
Modules 0 3 2016-06-21 0
MountPoints2 0 5 2016-06-21 0
NewShortcutHandlers 1 0 2016-06-21 0
Package Installation 1 0 2016-06-21 0
RecentDocs 6 4 2016-06-21 1
Ribbon 1 0 2016-06-21 1
RunMRU 0 0 2016-06-21 1
SearchPlatform 0 1 2016-06-21 0
Shell Folders 31 0 2016-06-21 0
Shutdown 1 0 2016-06-21 0
StartPage 2 0 2016-06-21 0
Values ComDlg32 OpenSavePidlMRU
Drag a column header here to group by that column
Extension Value Name Mru Position Absolute Path Opened On
" 1 0 My Computer\C:\Users\Hunter\Documents\Welcome.docx 2016-06-21 12:27:37
docx 0 0 My Computer\C:\Users\Hunter\Documents\Welcome.docx 2016-06-21 12:27:37
jpg 0 0 My Computer\Documents\home-network-design-networking-for-a-single-family-home-case-house-arkko-1433-x-792.jpg 2016-06-21 12:18:17
" 0 1 My Computer\Documents\home-network-design-networking-for-a-single-family-home-case-house-arkko-1433-x-792.jpg
Total rows: 4
Export
Type viewer ComDlg32 OpenSavePidlMRU selected row details
Type: Root folder: GUID, Value: My Computer
Type: Drive letter, Value: C:
Type: Directory, Value: Users
Extension blocks found: 1
Long name: Users
Localized name: @shell32.dll,-21813
Created: 8/22/2013 1:36:16 PM +00:00
Last access: 6/21/2016 8:37:46 AM +00:00
MFT entry/sequence #: 2121/1 (0x849/0x1)
File system hint: NTFS
Short name: Users
Modified: 6/21/2016 8:37:46 AM +00:00
Type: Directory, Value: Hunter
Extension blocks found: 1
Key: Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU
Value: None Collapse all hives
Selected hive: NTUSER.DAT Last write: 2016-06-21 12:27:37 Key contains no values Load complete
Hidden keys: 0

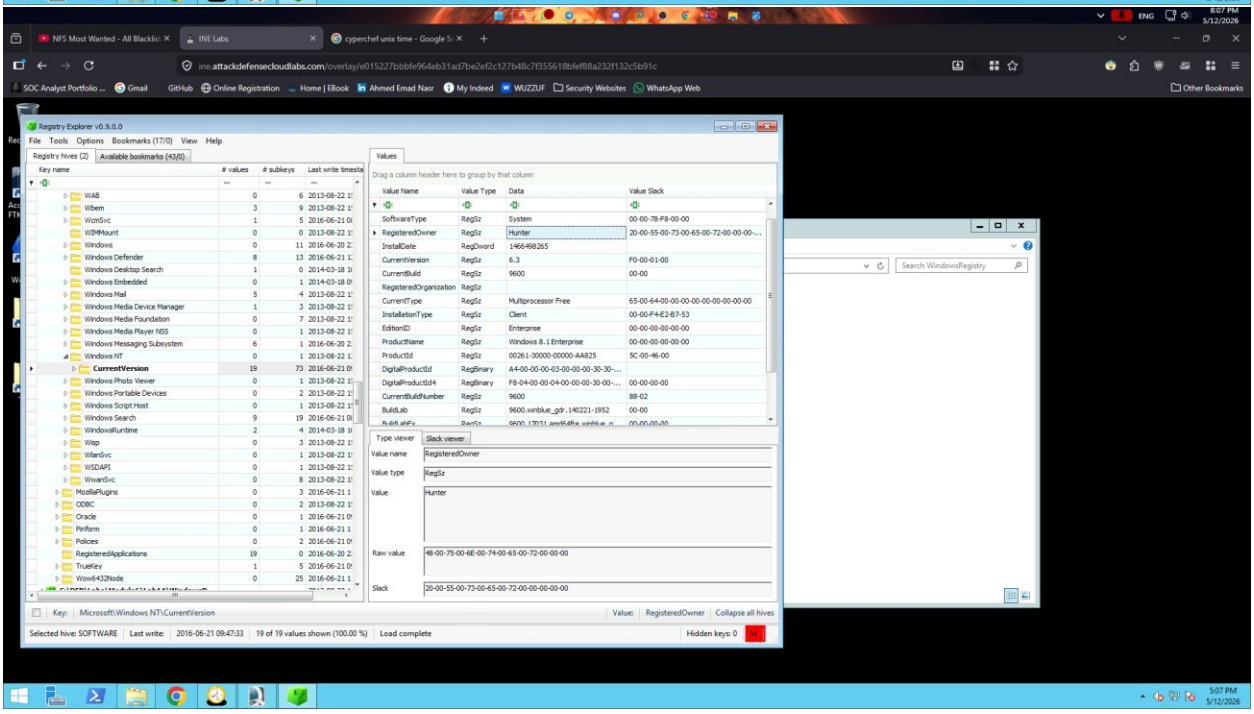

Registry Explorer v0.9.0.0
ConnectedSearch
Controls Folder (Wow64)
DeviceAccess
Explorer
Accent
Advanced
AppContract
AutoplayHandlers
BitBucket
CabinetState
CD Burning
CIDOpen
CIDSave
CLSID
ComDlg32
CIDSizeMRU
FirstFolder
LastVisitedPidlMRU
OpenSavePidlMRU
ControlPanel
Discardable
FileExts
HelpSticker
HideDesktopIcons
LowRegistry
MenuOrder
Modules
MountPoints2
NewShortcutHandlers
Package Installation
RecentDocs
Ribbon
RunMRU
SearchPlatform
Shell Folders
Shutdown
StartPage
1 0 {1946CF6E-0 1C9-43C7-877D-817884A047D6} My Computer\C:\Users\Hunter\Documents 2016-06-21 12:27:37
0 1 chrome.exe My Computer\Documents
01 00 00 00 00 00 00 00 FF FF FF FF
Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidlMRU
Selected hive: NTUSER.DAT
Last write: 2016-06-21 12:27:37
3 of 3 values shown (100.00 %)
Load complete

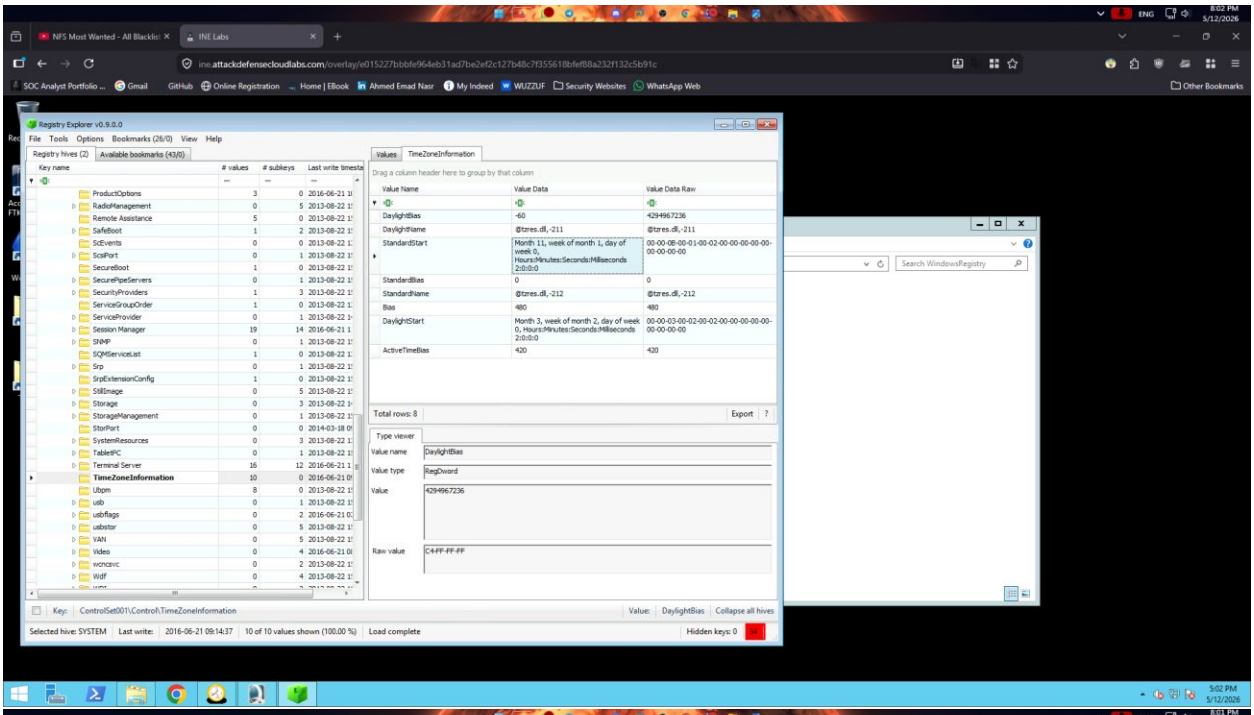

Registry Explorer v0.9.0.0
MMDevices
MSSHA
NcdAutoSetup
NetCache
NetworkServiceTriggers
OEMInformation
OneDrive
OOBE
OpenWith
OptimalLayout
Parental Controls
Personalization
PhotoManager
PhotoPropertyHandler
Policies
PrecisionTouchPad
Prelaunch
PreviewHandlers
PropertySystem
Proximity
PushNotifications
Reliability
RenameFiles
Run
RunOnce
SelectiveRemoteWipe
SettingSync
Setup
SharedDLLs
Shell
Shell Extensions
ShellCompatibility
ShellServiceObjectDelayLoad
SideBySide
SkyDriveRamps
SMDEn
SMI
VBoxTray RegSz C:\Windows\system32\VBoxTray.exe 00-20
C:\Windows\system32\VBoxTray.exe
43-00-3A-00-5C-00-57-00-69-00-6E-00-64-00-6F-00-77-00-73-00-5C-00-73-00-79-00-73-00-74-00-65-00-6D-00-33-00-32-00-5C-00-56-00-42-00-6F-00-78-00-54-00-72-00-61-00-79-00-2E-00-65-00-78-00-65-00-00-00
00-20
Microsoft\Windows\CurrentVersion\Run
Selected hive: SOFTWARE
Last write: 2016-06-21 09:47:24
1 of 1 values shown (100.00 %)
Load complete

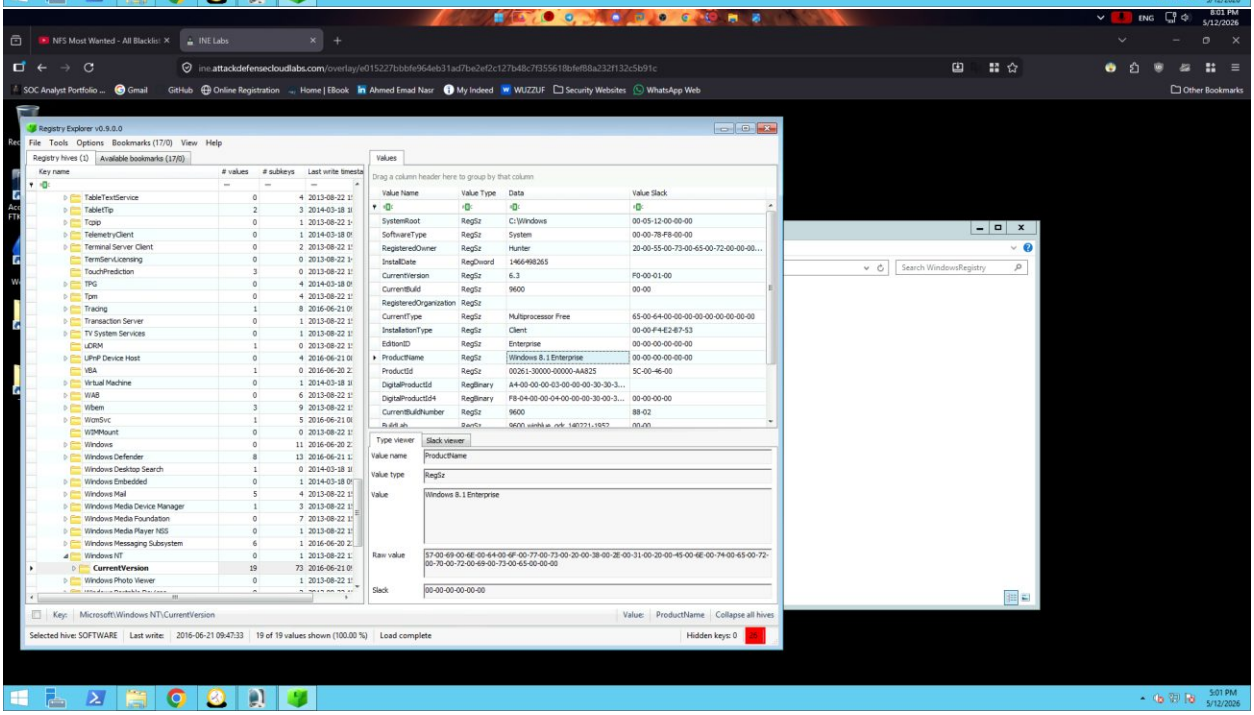

Registry Explorer v0.9.9.0
SnippingTool.exe
%SystemRoot%\system32\SnippingTool.exe
00-00-00-00-00-00


Registry Explorer v0.9.9.0
C:\DFP\Labs\Modules\Lab14\WindowsR...
SAM\Domains\Account\Users


Registry Explorer v0.9.0.0
File Tools Options Bookmarks (17/0) View Help
Registry hives (1) Available bookmarks (17/0)
Values
Value Name Value Type Data Value Slack
ShowTabletKeyboard RegDword 0
IdleTime RegDword 900 16
SelectedUserSID RegSz S-1-5-21-2489440558-2754304563-710705792-1001
LastLoggedOnUser RegSz Hunter
LastLoggedOnProvider RegSz {60B78E88-EAD8-445C-9CFD-0B87F74EA6CD}
NetworkStatusType RegDword 0
LastLoggedOnSAMUser RegSz .\Hunter
LastLoggedOnUserSID RegSz S-1-5-21-2489440558-2754304563-710705792-1001
Type viewer Slack viewer
Value name LastLoggedOnUser
Value type RegSz
Value Hunter
Raw value 48-00-75-00-6E-00-74-00-65-00-72-00-00-00
Slack 72-00-00-00-61-00-74-00-6F-00-72-00-00-00-72-00-00-00-31-36-33-38
Key: Microsoft\Windows\CurrentVersion\Authentication\LogonUI
Value: LastLoggedOnUser
Collapse all hives
Selected hive: SOFTWARE
Last write: 2016-06-21 01:42:40
8 of 8 values shown (100.00 %)
Load complete
Hidden keys: 0


Registry Explorer v0.9.0.0
File Tools Options Bookmarks (17/0) View Help
Registry hives (2) Available bookmarks (43/0)
Values
Value Name Value Type Data Value Slack
SoftwareType RegSz System
RegisteredOwner RegSz Hunter
InstallDate RegDword 1466498265
CurrentVersion RegSz 6.3
CurrentBuild RegSz 9600
RegisteredOrganization RegSz
CurrentType RegSz Multiprocessor Free
InstallationType RegSz Client
EditionID RegSz Enterprise
ProductName RegSz Windows 8.1 Enterprise
ProductId RegSz 00261-30000-00000-AA825
DigitalProductId RegBinary
DigitalProductId4 RegBinary
CurrentBuildNumber RegSz 9600
BuildLab RegSz 9600.winblue_gdr.140221-1952
Type viewer Slack viewer
Value name RegisteredOwner
Value type RegSz
Value Hunter
Raw value 48-00-75-00-6E-00-74-00-65-00-72-00-00-00
Slack 20-00-55-00-73-00-65-00-72-00-00-00-00-00
Key: Microsoft\Windows NT\CurrentVersion
Value: RegisteredOwner
Collapse all hives
Selected hive: SOFTWARE
Last write: 2016-06-21 09:47:33
19 of 19 values shown (100.00 %)
Load complete
Hidden keys: 0


NFS Most Wanted - All Blacklis:
INE Labs
ine.attackdefensecloudlabs.com/overlay/e015227bbbfe964eb31ad7be2ef2c127b48c7f355618bfef88a232f132c5b91c
Registry Explorer v0.9.0.0
File Tools Options Bookmarks (26/0) View Help
Registry hives (2) Available bookmarks (43/0)
Values TimeZoneInformation
Value Name Value Data Value Data Raw
DaylightBias -60 4294967236
DaylightName @tzres.dll,-211 @tzres.dll,-211
StandardStart Month 11, week of month 1, day of week 0, Hours:Minutes:Seconds:Milliseconds 2:0:0:0 00-00-08-00-01-00-02-00-00-00-00-00-00-00-00-00
StandardBias 0 0
StandardName @tzres.dll,-212 @tzres.dll,-212
Bias 480 480
DaylightStart Month 3, week of month 2, day of week 0, Hours:Minutes:Seconds:Milliseconds 2:0:0:0 00-00-03-00-02-00-02-00-00-00-00-00-00-00-00-00
ActiveTimeBias 420 420
Total rows: 8
Export
Type viewer
Value name DaylightBias
Value type RegDword
Value 4294967236
Raw value C4-FF-FF-FF
Key: ControlSet001\Control\TimeZoneInformation
Value: DaylightBias
Collapse all hives
Selected hive: SYSTEM
Last write: 2016-06-21 09:14:37
10 of 10 values shown (100.00 %)
Load complete
Hidden keys: 0
Search WindowsRegistry


NFS Most Wanted - All Blacklis:
INE Labs
ine.attackdefensecloudlabs.com/overlay/e015227bbbfe964eb31ad7be2ef2c127b48c7f355618bfef88a232f132c5b91c
Registry Explorer v0.9.0.0
File Tools Options Bookmarks (17/0) View Help
Registry hives (1) Available bookmarks (17/0)
Values
Value Name Value Type Data Value Slack
SystemRoot RegSz C:\Windows 00-05-12-00-00-00
SoftwareType RegSz System 00-00-78-F8-00-00
RegisteredOwner RegSz Hunter 20-00-55-00-73-00-65-00-72-00-00-00...
InstallDate RegDword 1466498265
CurrentVersion RegSz 6.3 F0-00-01-00
CurrentBuild RegSz 9600 00-00
RegisteredOrganization RegSz
CurrentType RegSz Multiprocessor Free 65-00-64-00-00-00-00-00-00-00-00-00
InstallationType RegSz Client 00-00-F4-E2-87-53
EditionID RegSz Enterprise 00-00-00-00-00-00
ProductName RegSz Windows 8.1 Enterprise 00-00-00-00-00-00
ProductId RegSz 00261-30000-00000-AA825 5C-00-46-00
DigitalProductId RegBinary A4-00-00-00-03-00-00-00-30-30-3...
DigitalProductId4 RegBinary F8-04-00-00-04-00-00-00-30-00-3... 00-00-00-00
CurrentBuildNumber RegSz 9600 88-02
Type viewer Slack viewer
Value name ProductName
Value type RegSz
Value Windows 8.1 Enterprise
Raw value 57-00-69-00-6E-00-64-00-6F-00-77-00-73-00-20-00-38-00-2E-00-31-00-20-00-45-00-6E-00-74-00-65-00-72-00-70-00-72-00-69-00-73-00-65-00-00-00
Slack 00-00-00-00-00-00
Key: Microsoft\Windows NT\CurrentVersion
Value: ProductName
Collapse all hives
Selected hive: SOFTWARE
Last write: 2016-06-21 09:47:33
19 of 19 values shown (100.00 %)
Load complete
Hidden keys: 0
Search WindowsRegistry